

Venkatasubramani Karthikeyan

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SUMMARY

Experienced Machine Learning engineer with 4 years of industry expertise, specializing in the retail domain. Possessing a strong command of Python, SQL, and machine learning algorithms. Eagerly seeking opportunities in data science to further enhance skills and contribute to impactful projects. Actively seeking data science opportunities to leverage acquired expertise and continue professional growth.

EDUCATION

The University of Leicester, Leicester, The UK (Sep 2022 – Nov 2023)

Master of Science (MSc) in Data Analysis for Business Intelligence

- Cleared with Distinction, 74%
- Modules Completed: Fundamentals of Data Science, Data Mining and Neural Networks, Analysis of Algorithms, Generalized Linear Models, Linear Statistical Models (Mathematics and Statistics), and Data Analytics for eSports.

Anna University - Sri Sairam Engineering College, Chennai, India (Jul 2014 - May 2018)

Bachelor of Engineering (B.E) in Electrical and Electronics Engineering

- CGPA: 7.89/10 (First Class)
- Relevant Modules: Object Oriented Programming (OOPS), Computer Programming, Transforms & Partial Differential Equations, Numerical methods.

WORK EXPERIENCE

Tata Consultancy Services, Chennai, India – Machine Learning Engineer (Jun 2018 – Aug 2022)

- Incorporated Retail data to understand and translate complex business problems into machine learning problems.
- Seasoned Machine Learning Engineer proficient in Python, Pyspark, and SQL, specializing in Optumera pricing and promotion optimization on Azure Databricks.
- Successfully optimized code runtime by migrating from pandas to Pyspark and SQL, achieving a remarkable threefold reduction in execution time.
- Proficient in version control systems like Git, with hands-on experience in utilizing CI/CD pipelines for continuous integration and delivery
- Conducted comprehensive analysis using Time Series algorithms (Sarimax, Sarima, STL, FBprophet, Kats, and Stldecompose) for accurate demand forecasting.
- Implemented diverse regression analysis techniques and advanced clustering approaches, ensuring precise unit price calculation and effective handling of unpredictable scenarios.
- Demonstrated versatility by contributing to front-end tasks involving HTML, CSS, and JS, fostering a comprehensive understanding of the application's ecosystem. Additionally, actively engaged in regular

client meetings, promoting effective collaboration, and consistently improving model accuracy from 30% to 90%.

- Facilitated knowledge transfer and provided comprehensive training to a couple of junior team members, enabling them to work autonomously before leaving the job.

Nebuli, London, United Kingdom – Graduate Student Researcher (Jul 2023 – Sep 2023) (Dissertation)

- Advanced Topic Classification for PubMed Research Articles: Developed and implemented an NLP-driven approach to automatically classify and group vast PubMed research articles into multiple relevant topics, enhancing article discoverability and interdisciplinary research potential.
- NLP Feature Engineering & Analysis: Leveraged techniques like 'Bag of Words' for feature extraction, performed in-depth statistical analysis of word frequencies, and employed both classification and clustering methodologies to discern and categorize core themes present in research articles.
- Benchmarking against Leading Language Models: Pioneered a model comparison framework to evaluate custom-built classifiers against renowned pre-trained models like BERT, BART, DeBERTa, and GPT-2, yielding insights into optimization strategies and establishing benchmarks for future research endeavours in collaboration with Nebuli Company.

University of Leicester, Leicester, United Kingdom – Student Intern (Oct 2022 – Feb 2023) (part-time)

- Conducted focused analysis on health, migration, and ethnicity aspects using census data, providing targeted insights in these areas.
- Employed a diverse toolkit including Excel, Python, and Plotly for data manipulation, analysis, and visualization, enabling comprehensive exploration and effective communication of findings.
- Initiated and facilitated data-focused conversations with key stakeholders involved in planning, service intervention, and practice, fostering informed decision-making and collaborative efforts for leveraging the insights gained from the census data.

PROJECTS

Chess.com Dashboard:

- Developed a personalized dashboard for chess.com using Python and the chess.com API.
- Implemented Plotly Dash framework for interactive visualizations, including rating timeseries, win-loss percentage, recent games, piece capture distribution, opponent country distribution, time/move analysis, and opening analysis.

Data Visualization:

- Created interactive visualizations using Plotly framework, including Sankey charts, Sunburst charts, Multiple Pie charts, horizontal bar charts, and Rating charts.
- Utilized interactivity features to enhance data exploration and analysis capabilities.

Mathematical Modelling using Covid data:

- Applied logistic growth, SIR, and Extended SIR models to COVID-19 data from Finland, Germany, and Italy.
- Selected the most accurate model based on multiple criteria to predict the spread and impact of the pandemic.

Stock market data analysis using Pyspark and Python:

- Conducted stock market data analysis using Pyspark and Python.

- Explored various time series algorithms, such as AR, MA, ARMA, SARIMA, SARIMAX, fbprophet, LSTM, for forecasting stock prices and capturing market trend.

SOFT SKILLS

- Communication
- Proactive team player
- Active Listening
- Public speaking

TECHNICAL SKILLS

- **Data Science Tools:** Python, SQL, Pandas, Scikit Learn, TensorFlow, MS Excel, Pyspark
- **Data Science Concepts:** Regression, Classification, Clustering, Decision Tree, XG Boost, Random Forest, Natural Language Processing (NLP) and Timeseries analysis
- **Data Science Environment:** Azure Databricks, Kaggle, Jupyter Notebook
- Proficient in basics of HTML, JavaScript, CSS.

LANGUAGES

- English
- Tamil

VOLUNTEERING

Data Club President (University of Leicester)

- Imparted knowledge to coding novices and those unfamiliar with industry practices.
- Collaborated in an agile framework.
- Successfully tackled and met problem statement objectives.

HOBBIES/INTERESTS

- Play chess
- Explore restaurants
- travel
- Hiking
- Reading advanced achievements in AI, Data Science and ML. Such as Large Language Models, Generative AI, Chatgpt etc.

ACCOMPLISHMENTS

- Awarded best outgoing student in college.
- Service commitment award in Tata Consultancy Services.